Finance Research Letters 58 (2023) 104464

Contents lists available at ScienceDirect

Finance Research Letters

journal homepage: www.elsevier.com/locate/frl

Optimal capital investment structure and sustainable development: Evidence from China

Yiyang Cheng [a,*], Shaofeng Ru [a,b], Gavin Kader [c]

[a] Economics and Management Department of Northwest University, Xi'an, 710127, Shaanxi, China
[b] The Western Economic Development Research Institute, Northwest University, Xi'an, 710127, Shaanxi, China
[c] Southwestern University of Finance and Economics, Chengdu, 611130, Sichuan, China

ARTICLE INFO

Keywords:
Sustainable development
Capital structure
Human capital
Environmental capital
Social capital
Deviation degree

ABSTRACT

This paper studies how capital structure (physical, human, environmental, and social capital) affects sustainable development. This paper constructs a model for optimal sustainable development and deduces the necessary condition is that the input ratio of per capita human capital, environmental capital, and social capital are equal to their respective output elasticity ratios. Using data from China (2007–2020), empirical tests confirm smaller deviations between output elasticity ratios and the actual input ratio are associated with higher sustainable development. This paper concludes that, unlike physical capital, investment in human, environmental, and social capital promote sustainable development.

1. Introduction

The 2022 Sustainable Development Report by the United Nations highlights that challenges such as the climate crisis, COVID-19, and global conflicts jeopardize the achievement of the 17 Sustainable Development Goals (Sachs et al., 2022). Echoing this concern, China had previously adopted a proactive and adaptive approach in pursuit of sustainable development. In October 2015, China introduced a new development concept focusing on 5 different concepts: innovation, coordination, green, openness, and sharing. Key strategies encompass deepening supply-side structural reforms, fostering innovation, rural revitalization, regional coordinated development, and comprehensive international engagement.[1]

By December 2016, China responded to the United Nations '2030 Agenda for Sustainable Development' by launching the *'Plan for the Construction of Innovative Demonstration Zones for Implementing the 2030 Agenda for Sustainable Development'*.[2] In September 2020, China made a significant environmental commitment by setting goals to peak carbon emissions by 2030 and achieve carbon neutrality by 2060.[3] Since 2021, the Chinese government has initiated measures to curb unchecked capital expansion, aiming to prevent excessive profit-driven activities that could harm the environment and societal well-being. Historically, China's economic growth was heavily reliant on physical capital investment, with limited emphasis on newer forms of capital (Wang and Li, 2022). This growth, fueled by the exploitation of natural resources and inexpensive labor, often diverged from sustainable development

* Corresponding author.
E-mail addresses: chengyiyang@stumail.nwu.edu.cn (Y. Cheng), rsf00@nwu.edu.cn (S. Ru), kadergavin@swufe.edu.cn (G. Kader).

[1] This development concept was introduced during the Fifth Plenum of the 18th Central Committee of the Communist Party of China in October 2015.

[2] Released by the State Council on December 3, 2016, this plan aims to achieve global development targets spanning poverty alleviation, health, education, and environmental conservation over the next 15 years, all underpinned by sustainable development principles.

[3] Announced at the 75th United Nations General Assembly in September 2020, these targets underscore China's ambition to harmonize economic progress with ecological responsibility.

https://doi.org/10.1016/j.frl.2023.104464
Received 12 June 2023; Received in revised form 30 August 2023; Accepted 10 September 2023
Available online 16 September 2023

principles. Contrasting with the traditional growth paradigm, sustainable development advocates for a holistic investment approach, balancing physical, human, environmental, and social capital.

Stern et al. (2020) emphasize in their report titled *'Valuing and investing in physical, human, natural, and social capital in the 14th Plan'* that China ought to focus on enhancing industrial production efficiency, addressing its aging population challenges, improving air quality, and mitigating social inequality. Achieving sustainable development cannot be entrusted solely to either the government or the market. An exclusive reliance on the market might lead to the neglect of environmental conservation and social welfare, while depending solely on the government could compromise economic efficiency. Consequently, this study aims to strike a balance by considering both governmental and private sector roles. The objective is to discern an optimal investment strategy across the four capital types, facilitating a shift in the economic development paradigm, addressing market failures, and ultimately realizing sustainable development.

Sustainable development is multifaceted in its definition. From an environmental perspective, safeguarding natural resources is paramount for sustainable development and sustaining all forms of life (World Wildlife Fund, 1980) whereas a social perspective focus on elevating quality of human life, ensuring a minimum level is maintained within the capacity of the ecosystem (Munro, 1991). The economics perspective entails maximizing the net benefits of economic progress while preserving the integrity of natural resources and their associated services (Barbier, 2013). Notice how this is holistic, intertwining environmental, social, and economic dimensions (Giddings et al., 2002). Thus, sustainable development seeks inclusive growth, harmonizing both quantitative and qualitative economic growth. It still champions economic growth without sidelining it for environmental conservation. As such, sustainable development essentially mandates a balanced consideration of economic expansion, quality of life, environmental stewardship, and social equity. Based on these insights, this study has formulated a model for sustainable development.

The model's theoretical foundation draws upon (Ekins et al., 2008), who introduces a four-capital framework to analyze the roles of physical, human, natural (environmental), and social capital in promoting sustainable development. This framework diverges from traditional economic growth models, which probjedominantly focus on the contributions of physical or human capital to GDP or per capita income, typically incorporating a purely consumption-related utility function. In contrast, sustainable development encompasses not just growth, but also utility derived from human, environmental, and social equity dimensions. In “Quality of Growth”, Thomas (2000) emphasizes that holistic improvements in social welfare are contingent upon adequate investments in physical, human, and natural capital. Stern et al. (2020) further underscores the significance of these four capital types as pivotal investment avenues for China between 2021 and 2025. Building on this theoretical foundation, this paper enhances conventional economic growth models, moving beyond a solely consumption-based utility function to include components of utility associated with human, environmental, and social capital.

While the aforementioned theory provides an analytical and qualitative framework for sustainable development, it remains silent on the optimal proportions for investing in different types of capital. Much of the existing literature tends to examine the impact of individual capital types on sustainable development in isolation, with little discussion on a comprehensive model that integrates different forms of capital. A key contribution of this paper is the introduction of a novel model to analyze the optimal allocation of these different forms of capital to realize sustainable development. Clearly, this holds significant implications for investment strategies. Empirically, this paper constructs an indicator to measure sustainable development encompassing the previous background and theoretical discussions. Furthermore, data on sources of investment for each type of capital is distinguished between private and public sectors. Finally, by employing the translog function, this study computes the output elasticity of capital in China from 2007 to 2020, demonstrating that a well-balanced capital structure correlates with higher levels of sustainable development — a finding with profound policy implications. A diagrammatic representation of the framework of capital structure and sustainable development is shown in Fig. 1:

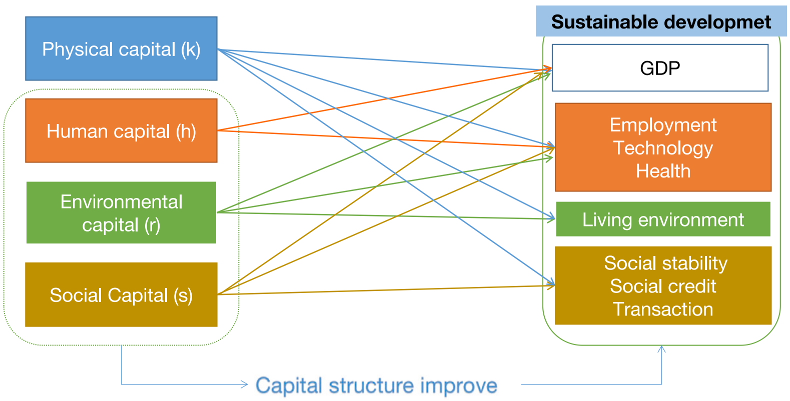

Fig. 1. The framework of capital structure and sustainable development.

2. Literature review

In the context of sustainable development, physical capital, as well as human capital, environmental capital, and social capital have all played significant roles.

Human Capital This encompasses education, training, medical care, and other aspects of knowledge and health enhancement (Becker, 1992). Public awareness, combined with education and training, is essential in guiding society towards sustainability (McKeown et al., 2002). The Sustainable Development Goals further emphasize this, particularly highlighting quality education in Goal 4, which underscores the vital importance of human capital (Sachs et al., 2022).

Environmental capital Similar to other capital forms, nature environment is typically long-lasting, providing services over extended periods and depreciating (Fisher, 1904). Investments primarily target pollution control, and maintenance of ecological balance. Policymakers are consistently encouraged to direct more investments into environmental technologies, aiming for enhanced economic outcomes that align with sustainability (Ulucak et al., 2020). Zhang and Wen (2008) advocate for China's commitment to national environmental protection strategies as a pathway to sustainable development.

Social capital Defined by the quality of interpersonal relationships, social networks, and levels of trust, trust is viewed as an essential component of social capital (Putnam, 2015). Various studies, such as Kusakabe (2012), discusses the positive impacts of diverse social capital accumulation on sustainable development. Helliwell et al. (2014) observe that communities with heightened social engagement levels tend to be happier than otherwise equivalent communities.

Physical capital Serving as the traditional cornerstone for economic growth, physical capital includes machinery, equipment, infrastructure, transportation and more. The accumulation of other forms of capital is contingent upon a base-level of physical capital.

While much of the existing literature focuses on individual capital types, an exploration of the interrelationships among the forms of capital is somewhat lacking. Identifying the optimal capital investment portfolio is essential to achieve and bolster sustainable development (Yoshino et al., 2021). Studies such as Çakar et al. (2021) investigate the relationship between human capital and environmental conservation, suggesting that increases in human capital can lead to innovations that mitigate environmental degradation. Meanwhile, Li et al. (2022) examine the dynamics between social indicators and environmental degradation, deducing that income inequality can exacerbate carbon emissions. Iram et al. (2020) discuss the equilibrium between environmental performance and economic growth.

Collectively, the literature underscores that the structural relationship between the different forms of capital plays a crucial role in sustainable development. To offer a rigorous insight into the interactions between different capital forms and sustainable development, this paper estimates a model which shows the necessity for balanced growth across the different forms of capital.

3. Model

3.1. Objective function

As previously discussed, based on Thomas (2000) and Ekins et al. (2008), the model uses a utility function which combines traditional consumption utility with a human, environmental and social utility. Assuming that society is composed of N individuals, the sum of the utility of each individual constitutes the total social utility, at any given point in time. The (additive) social welfare function for a society of N individuals (indexed by i) is expressed as follows:

$$U = \sum_{i=1}^{N} u(c_i) + \sum_{i=1}^{N} v(h_i, s_i, r_i) \tag{1}$$

The social welfare function is an increasing and concave function of consumption (c_i), human capital (h_i), environmental capital (r_i), and social capital (s_i).

In order to define the main objective function of the model, the first step is to take a Taylor expansion. This is because the model will take the form of a *dynamic* optimization problem with the social welfare function being identical across time. The (2nd order) Taylor expansion of U at consumption per capita, $\bar{c}$, human capital per capita capital, $\bar{h}$, environmental capital per capita, $\bar{r}$, and social capital per capita, $\bar{s}$ is given by:

$$\begin{aligned}
U \approx\ & N u(\bar{c}) + \sum_{i=1}^{N} u'(\bar{c})(c_i - \bar{c}) + \frac{1}{2}\sum_{i=1}^{N} u''(\bar{c})(c_i - \bar{c})^2 + N v(\bar{h}, \bar{s}, \bar{r}) + \sum_{i=1}^{N} v_h(\bar{h}, \bar{s}, \bar{r})(h_i - \bar{h}) \\
& + \sum_{i=1}^{N} v_s(\bar{h}, \bar{s}, \bar{r})(s_i - \bar{s}) + \sum_{i=1}^{N} v_r(\bar{h}, \bar{s}, \bar{r})(r_i - \bar{r}) + \frac{1}{2}\sum_{i=1}^{N} v_{hh}(\bar{h}, \bar{s}, \bar{r})(h_i - \bar{h})^2 \\
& + \frac{1}{2}\sum_{i=1}^{N} v_{ss}(\bar{h}, \bar{s}, \bar{r})(s_i - \bar{s})^2 + \frac{1}{2}\sum_{i=1}^{N} v_{rr}(\bar{h}, \bar{s}, \bar{r})(r_i - \bar{r})^2 + \sum_{i=1}^{N} v_{hs}(\bar{h}, \bar{s}, \bar{r})(h_i - \bar{h})(s_i - \bar{s}) \\
& + \sum_{i=1}^{N} v_{hr}(\bar{h}, \bar{s}, \bar{r})(h_i - \bar{h})(r_i - \bar{r}) + \sum_{i=1}^{N} v_{sr}(\bar{h}, \bar{s}, \bar{r})(s_i - \bar{s})(r_i - \bar{r})
\end{aligned} \tag{2}$$

where $u'(\bar{c})$ and $u''(\bar{c})$ are the 1st and 2nd derivatives of u at $\bar{c}$, respectively. $v_h(\bar{h}, \bar{s}, \bar{r})$ and $v_{hh}(\bar{h}, \bar{s}, \bar{r})$ are the 1st and 2nd order partial derivatives of v with respect to human capital per capita, respectively. $v_r(\bar{h}, \bar{s}, \bar{r})$, $v_{rr}(\bar{h}, \bar{s}, \bar{r})$, $v_s(\bar{h}, \bar{s}, \bar{r})$ and $v_{ss}(\bar{h}, \bar{s}, \bar{r})$ are defined analogously for environmental and social capital. $v_{xy}(\bar{h}, \bar{s}, \bar{r})$ denotes the cross partial derivative with respect to x, y, with x and y denoting any two different forms of capital.

The above Taylor expansion is a long expression which can be simplified in two different ways. For the first set of simplifications, notice that the follow terms reduce to 0:

$$\sum_{i=1}^{N} u'(\bar{c})(c_i - \bar{c}) = u'(c)\sum_{i=1}^{N}(c_i - \bar{c}) = u'(\bar{c})\left(\sum_{i=1}^{N} c_i - \sum_{i=1}^{N} \bar{c}\right) = 0 \tag{3}$$

This is also true for human capital, environmental capital, and social capital:

$$\sum_{i=1}^{N} v_h(\bar{h},\bar{s},\bar{r})(h_i - \bar{h}) = \sum_{i=1}^{N} v_s(\bar{h},\bar{s},\bar{r})(s_i - \bar{s}) = \sum_{i=1}^{N} v_r(\bar{h},\bar{s},\bar{r})(r_i - \bar{r}) = 0 \tag{4}$$

For the second set of simplifications, let $N\sigma_x^2$ denote $\sum_{i=1}^{N}(x_i - \bar{x})^2$ and $N\sigma_{ab}$ denote $\sum_{i=1}^{N}(a_i - \bar{a})(b_i - \bar{b})$. Let $E(U)$ denote the average utility across individuals (dividing by N). Using both sets of simplifications, equation (2) can now be simplified down to:

$$\begin{aligned} E(U) = {} & u(\bar{c}) + \frac{1}{2}u''(\bar{c})\sigma_c^2 + v(\bar{h},\bar{s},\bar{r}) + \frac{1}{2}v_{hh}(\bar{h},\bar{s},\bar{r})\sigma_h^2 + \frac{1}{2}v_{ss}(\bar{h},\bar{s},\bar{r})\sigma_s^2 + \frac{1}{2}v_{rr}(\bar{h},\bar{s},\bar{r})\sigma_r^2 \\ & + \sum_{i=1}^{N} v_{hs}(\bar{h},\bar{s},\bar{r})\sigma_{hs} + \sum_{i=1}^{N} v_{hr}(\bar{h},\bar{s},\bar{r})\sigma_{hr} + \sum_{i=1}^{N} v_{sr}(\bar{h},\bar{s},\bar{r})\sigma_{sr} \end{aligned} \tag{5}$$

Assuming that the choices of investment between the types of capital are independent of each other i.e. $v_{hs}(\bar{h},\bar{s},\bar{r}) = v_{hr}(\bar{h},\bar{s},\bar{r}) = v_{sr}(\bar{h},\bar{s},\bar{r}) = 0$, equation (5) reduces to:

$$E(U) = u(\bar{c}) + \frac{1}{2}u''(\bar{c})\sigma_c^2 + v(\bar{h},\bar{s},\bar{r}) + \frac{1}{2}v_{hh}(\bar{h},\bar{s},\bar{r})\sigma_h^2 + \frac{1}{2}v_{ss}(\bar{h},\bar{s},\bar{r})\sigma_s^2 + \frac{1}{2}v_{rr}(\bar{h},\bar{s},\bar{r})\sigma_r^2 \tag{6}$$

For the assumption, in the empirical analysis, investments in different forms of capital can be isolated from each other. For instance, investments in environmental infrastructure can be disentangled from the physical capital account, aligning with the model's independence assumption. Moreover, investment decisions in physical/environmental capital are inherently independent of other forms. This is underscored by the fact that these investments can be empirically isolated, suggesting that the utility derived from each can be considered distinct. The assumption of independence between social and human capital warrants further justification. This paper posits that when it comes to decision-making, these forms of capital are separable. Enhancements in human capital typically involve individuals refining their personal skills, while social capital pertains to their interactions with others, representing a different skill set. Within the model, current investments in specific capital forms can be substituted with others, given budgetary constraints. In this sense, an individual might prioritize human capital now, intending to compensate with investments in social skills later. However, this relationship is perceived as being *indirect* through the budget set, rather than as direct trade-offs. For instance, accumulating human capital might yield positive spillovers for other capital forms. The model can account for this by adjusting future investments, potentially reducing allocations to human capital in favor of other forms.

The study uses this social welfare function as the main objective function of the model. By concavity, the total welfare function is an increasing function of $\bar{c}$, $\bar{h}$, $\bar{s}$, $\bar{r}$ and a decreasing function of σ_c^2, σ_h^2, σ_s^2, and σ_r^2. In words, when the per capita consumption gap, per capita human capital gap, per capita social capital gap, and/or per capita environmental capital gap decrease, the total welfare increases. Further assume that σ_c^2, σ_h^2, σ_s^2, σ_r^2 are all zero i.e. on average, individuals are not deviating from average levels of consumption, human capital, environmental capital, and social capital.[4] The objective function to be maximized is given by:

$$\int_0^{\infty} E(U)e^{-\rho t}dt = \int_0^{\infty} \left[u(\bar{c}) + v(\bar{h},\bar{s},\bar{r})\right] e^{-\rho t}dt \tag{7}$$

where ρ is the discount factor and the functional form for $u(\cdot)$ is given by a constant relative risk aversion (CRRA) utility function[5]:

$$u(\bar{c}) = \frac{\bar{c}^{1-\theta} - 1}{1-\theta}, \quad 0 < \theta < 1 \tag{8}$$

where θ represents the coefficient of relative risk aversion. Assume that the marginal utility of $\bar{h}$, $\bar{s}$, $\bar{r}$ are constant and identical, denoted by ω. Let $v(\bar{h},\bar{s},\bar{r}) = w(\bar{h} + \bar{s} + \bar{r})$.

3.2. Optimization problem

Suppose that the sources of capital are provided by the central government the and private sector. I_X^g is the government's investment in $X \in \{H, R, S\}$, with H, R, S representing human, environmental, and social capital per capita. I_Y^p is the private sector's investment in $Y \in \{H, S\}$.[6] The capital accumulation equations for each form of capital per capita is as follows:

$$\dot{\bar{k}} = \bar{y} - \bar{c} - I_H^g - I_H^p - I_R^g - I_S^g - I_S^p \tag{9}$$

[4] By definition, large and unexpected deviations from average behavior will lead to welfare loss. Thus, the model assumes that individuals do not want to deviate from average behavior meaning the optimal capital portfolio can be determined without needing to account for this kind of welfare loss.

[5] The CRRA utility function can be used to describe the choices made by individuals and families in the face of uncertainty.

[6] The private sector does not invest in environmental capital. This is supported by the fact that the private sector invests negligibly in environmental capital compared to the government.

$$\dot{\overline{h}} = I_H^g + I_H^p \tag{10}$$

$$\dot{\overline{r}} = I_R^g \tag{11}$$

$$\dot{\overline{s}} = I_S^g + I_S^p \tag{12}$$

where $\dot{\overline{k}}, \dot{\overline{h}}, \dot{\overline{r}}, \dot{\overline{s}}$ denote the derivative of physical, human, environmental, and social capital per capita, respectively.

Assume a Cobb–Douglas functional form for the production function:

$$\overline{y} = A\left(\overline{k}\right)^{\alpha}\left(\overline{h}\right)^{\beta}\left(\overline{r}\right)^{\gamma}\left(\overline{s}\right)^{1-\alpha-\beta-\gamma} \qquad (0 < \alpha, \gamma, \beta < 1) \tag{13}$$

$\alpha, \beta, \gamma, 1-\alpha-\beta-\gamma$, and A are the output elasticities of physical capital per capita, human capital per capita, social capital per capita, environmental capital per capita, and total factor productivity (constant), respectively.

Assume that $\overline{c}_{min}$ represents the minimum per capita consumption required to sustain an individual's living standard. Consequently, individual per capita consumption should exceed this minimum threshold, implying $\overline{c} > \overline{c}_{min}$.

To derive optimal sustainable development in terms of the different forms of capital and consumption is to solve the following optimization problem:

$$\max \int_0^{+\infty} [u(\overline{c}) + v(\overline{h}, \overline{s}, \overline{r})] e^{-\rho t} dt \tag{14}$$

$$s.t. \begin{cases} \dot{\overline{k}} = \overline{y} - \overline{c} - I_H^g - I_H^p - I_R^g - I_S^g - I_S^p \\ \dot{\overline{h}} = I_H^g + I_H^p \\ \dot{\overline{r}} = I_R^g \\ \dot{\overline{s}} = I_S^g + I_S^p \\ \overline{c} > \overline{c}_{min} \end{cases} \tag{15}$$

3.3. Solving for the optimal path

As $u(\overline{c}) + v(\overline{h}, \overline{r}, \overline{s})$ is positive and bounded, the objective function converges. The corresponding Hamiltonian function is:

$$\begin{aligned} H = {} & u(\overline{c}) + v(\overline{h}, \overline{s}, \overline{r}) + \lambda(\bar{y} - \bar{c} - I_H^g - I_H^p - I_R^g - I_S^g - I_S^p) + m(I_H^g + I_H^p) + nI_R^g + l(I_S^g + I_S^p) \\ & + z(\overline{c} - \overline{c}_{min}) \end{aligned} \tag{16}$$

Solving the first-order necessary conditions to maximize total utility (see appendix) yields two necessary conditions:

$$\frac{\overline{h}^*}{\beta} = \frac{\overline{r}^*}{\gamma} = \frac{\overline{s}^*}{1-\alpha-\beta-\gamma} \tag{17}$$

$$\overline{c}^* = \left[A\left(\frac{\gamma}{\beta}\right)^{\gamma}\left(\frac{1-\alpha-\beta-\gamma}{\beta}\right)^{1-\alpha-\beta-\gamma} \frac{\alpha - \beta\frac{\overline{k}^*}{\overline{h}^*}}{\omega\left(\frac{\overline{k}^*}{\overline{h}^*}\right)^{1-\alpha}} \right]^{\frac{1}{\theta}} \tag{18}$$

3.3.1. Discussion of the necessary conditions

From a deeper examination of the model's necessary conditions, three major insights emerge: (1) To maximize sustainable development, the ratio between any two – whether human capital per capita, environmental capital per capita, or social capital per capita – should match the ratio of their respective output elasticities, (2) at the point of optimal sustainable development, if consumption is low/high, the ideal level of physical capital should be correspondingly high/low relative to human capital, environmental capital, and/or social capital and, (3) relying solely on increasing physical capital investment might result in a decline in sustainable development. Moreover, as consumption increases, maximizing sustainable development necessitates a further emphasis on human capital, environmental capital, and social capital. These points are further explicated below:

1. From Eq. (17), notice that, on the optimal path:
   (a) Human capital per capita, environmental capital per capita and social capital per capita are proportional to each other. Further, there is a positive relationship between each of them.
   (b) The ratio of any two of human capital per capita, environmental capital per capita or social capital per capita should be equal to the ratio of their respective output elasticities.

2. For exposition, re-write Eq. (18) as

$$\bar{c}^{\theta}\omega\left(\frac{\bar{k}^*}{\bar{h}^*}\right)^{1-\alpha}+T\beta\left(\frac{\bar{k}^*}{\bar{h}^*}\right)=T\alpha \tag{19}$$

where

$$T=A\left(\frac{\gamma}{\beta}\right)^{\gamma}\left(\frac{1-\alpha-\beta-\gamma}{\beta}\right)^{1-\alpha-\beta-\gamma}$$

Given T, the ratio of physical capital per capita to human capital per capita is a function of consumption per capita. When consumption per capita increases, the ratio of physical capital per capita to human capital per capita decreases. When sustainable development is maximized, $\left(\bar{k}^*/\bar{h}^*\right)$ is (non-linearly and) negatively correlated with per capita consumption $\bar{c}^*$. As previously mentioned, there is a positive relationship between per capita human capital, per capita environmental capital, and per capita social capital. This implies that $\left(\bar{k}^*/\bar{r}^*\right)$, $\left(\bar{k}^*/\bar{s}^*\right)$ and $\bar{c}^*$ are also negatively correlated with consumption per capita consumption. Consequently, at lower consumption levels, investments should be more focused in physical capital. However, at higher consumption levels, investments in human capital, environmental capital, and social capital should take priority.

3. Substituting $\bar{c}^*$, $\bar{h}^*$, $\bar{r}^*$ and $\bar{s}^*$ into the objective function yields the following value function i.e. the maximal social welfare function:

$$E(U^*)=u(\bar{c}^*)+\upsilon(\bar{h}^*,\bar{s}^*,\bar{r}^*) \tag{20}$$

Using the first-order conditions yields,[7]:

$$E(U^*)=u\left(\frac{\bar{k}^*}{\bar{h}^*}\right)+\upsilon(\bar{h}^*) \tag{21}$$

Fixing $\bar{h}^*$, let $\bar{k}'>\bar{k}^*$, then $\frac{\bar{k}^*}{\bar{h}'}>\frac{\bar{k}^*}{\bar{h}^*}$ and $\bar{c}'<\bar{c}^*$. Similarly, $u\left(\frac{\bar{k}'}{\bar{h}^*}\right)<u\left(\frac{\bar{k}^*}{\bar{h}^*}\right)$ with $\upsilon(\bar{h}^*,\bar{s}^*,\bar{r}^*)$ constant, then $E(U')<E(U^*)$. In this case, the level of sustainable development declines. In words, when fixing the level of human/environmental/social capital, increases in physical capital worsen sustainable development.

Now, fixing $\bar{k}^*$, let $\bar{h}'>\bar{h}^*$, then $\frac{\bar{k}^*}{\bar{h}'}<\frac{\bar{k}^*}{\bar{h}^*}$ and $\bar{c}'>\bar{c}^*$. As such, $u\left(\frac{\bar{k}^*}{\bar{h}'}\right)>u\left(\frac{\bar{k}^*}{\bar{h}^*}\right)$ and $\upsilon(\bar{h}',\bar{s}^*,\bar{r}^*)>\upsilon(\bar{h}^*,\bar{s}^*,\bar{r}^*)$, meaning $E(U')>E(U^*)$ i.e. the level of sustainable development improves. The above works similarly with environmental capital and social capital. In words, when fixing the level of physical capital, increases in human/environmental/social capital improve sustainable development.

Based on the analysis presented above, the following two empirical research hypotheses are proposed:

H1. When the ratio of any two of human capital per capita, environmental capital per capita or social capital per capita are closer to their estimated ratio of their respective output elasticities, sustainable development will be higher.

H2. The ratio of physical capital per capita with each of human capital per capita, environmental capital per capita, and social capital per capita has a non-linear and inverse relationship with per capita consumption i.e. $(k/h),(k/r),(k/s)$ have a non-linear and inverse relationship with c.

4. Empirical analysis

4.1. Data

4.1.1. Sustainable development indicator

Based on the four-capital model and the 17 Sustainable Development Goals (Sachs et al., 2022), this paper identifies four core themes to gauge sustainable development: economic growth and distribution, national quality, living standards, and market infrastructure. Investment in human capital can foster economic growth and enhance the population's quality of life. Environmental investments can elevate living standards, particularly the quality of the living environment. Allocating resources to social capital can mitigate social inequality, encourage social trust, and increase the volume of market transactions. While physical capital does contribute to sustainable development, reliance solely on it may prove insufficient (although necessary), especially when faced with diminishing marginal returns.

A comprehensive indicator of sustainable development is constructed based on the entropy-TOPSIS method from 2007 to 2020, with a total of 30 provinces from China.[8,9] (Table 1).

[7] The first-order conditions are written explicitly in the appendix in Eqs. (27) (28) and (29).

[8] Due to the unavailability of data, Tibet is excluded from the analysis.

[9] Entropy-TOPSIS method is used to solve multi-objective decision-making problems using the Euclidean distance function (Bhowmik et al., 2020). It is also a method be used to establish comprehensive evaluation indicators (Sun et al., 2017). The method is to first use the entropy weight method to calculate the weights of each indicator, and then rank the similarity between the evaluated object and the ideal solution.

Table 1
Multiple-index system of sustainable development.

Primary layer	Secondary layer	Third level layer	Indicator definition
Sustainable development	Economic growth and distribution	Per capita consumption expenditure (+)	Daily living expenses of each resident
		Per Capita GDP (+)	Ratio of actual GDP to total population
		Theil index (−)	Income gap between regions
		Urban-rural income ratio (−)	Income gap between urban and rural areas
	National quality	Average length of education (+)	The average of the total number of years of academic education
		Unemployment rate (−)	Ratio of unemployed population to working population
		Resident mortality rate (−)	Ratio of fatalities to total population
		Number of beds per capita (+)	Hospital beds owned by each person
		Technical transaction volume per capita (+)	Ratio of technology market turnover to total population
		Number of valid patents per capita (+)	Ratio of valid patents to total population
	Living standards	Green space area per capita (+)	The area of public green space occupied by each person
		Domestic waste treatment rate (+)	Ratio of domestic waste treatment capacity to waste output
		Exhaust emissions per capita (−)	Ratio of sulfur dioxide emissions to population
		Wastewater discharge per capita (−)	Ratio of wastewater discharge to population
	Market infrastructure	Market transaction volume exceeding 100 million yuan per capita (+)	Ratio of over 100 million market transactions to population
		Total telecommunications business volume services per capita (+)	Ratio of total telecommunications business volume to population
		Passenger volume per capita (+)	Passenger volume/total population
		Marketization process index[a](+)	Reflect the degree of marketization

Note: "+" in the brackets indicates that the indicator is positive, and "−" indicates that the indicator is negative. The above data is sourced from the National Bureau of Statistics of China, China Environmental Statistics Yearbook, China Social Statistics Yearbook, China Education Statistics Yearbook, China Health Statistics Yearbook, and China Science and Technology Statistics Yearbook.

[a] The calculation method of marketization index refers to Fan et al. (2011),which reflects the progress of marketization from five aspects: the relationship between the government and the market, the development of non-state-owned economy, the development level of product market, the development level of factor market, the development of market intermediary organizations, and the legal system environment.

4.1.2. Explanatory variables

To measure whether the investment ratios are equal to their respective output elasticity ratios, this study builds a deviation index drawing on the capital misallocation index of Hsieh and Klenow (2009).

Let D_{hr}, D_{hs}, D_{sr} measure proximity of investment (in capital) ratios to their respective elasticity ratio. e_h, e_r, e_s refer to the output elasticity of human capital, environmental capital, and social capital. τ_{hr}, τ_{hs}, τ_{sr} denote the ratio between the ratio of capital output elasticities and the (observed) investment ratio. The calculation method is as follows:

$$D_{xy} = \mid \tau_{xy} - 1 \mid \qquad \tau_{xy} = \frac{e_x}{e_y} / \frac{x}{y} \tag{22}$$

for $x, y \in \{h, s, r\}$ and $x \neq y$.

When τ_{hr}, τ_{hs}, τ_{sr} are 1, this means that D_{hr}, D_{hs}, D_{sr} are zero meaning no deviation. This is to say that when the ratio of capital output elasticities and the actual investment ratio are equal, there is no misallocation of investment in capital in terms of maximizing sustainable development.

The standard perpetual inventory formula is utilized to calculate measures of the four kinds of capital, with 2000 set as the base year The calculation method for the base period is to divide the investment in the base year by the sum of the average growth rate and depreciation rate of each province during the sample period (Hall and Jones, 1999). Table 2 summarizes the investment resources per form of capital.

Physical Capital: Investment in physical capital is measured by fixed capital investment. The depreciation rate is set to be 0.096 (Zhang, 2008). This paper uses the fixed assets investment price index to deflate the nominal investment with 2000 as the base year.

Human Capital: Human capital includes education, training, medical care, and other additions to knowledge and health (Flabbi and Gatti, 2018). Consistent with physical capital, this paper calculates human capital by a cost-based method. This paper calculates a measure of human capital assuming that the sources of human capital are from the government and the private sector. Public human capital investment can be calculated from the government's financial expenditure on education and health care. Revenue from education expenses (including tuition and miscellaneous expenses) represents the private sector's expenditure on education while residents' healthcare expenditure represents private investment in health. The depreciation rate of human capital of China is 0.047 (Zhang, 2020). Finally, this paper uses the consumer price index as the deflator.

Environmental Capital: This study uses public environmental protection expenditure and investment in urban environmental infrastructure to represent environmental capital with investment only coming from the government (consistent with modeling assumptions). Although there is no unified definition of the depreciation rate of environmental capital, Cobb and Daly (1989) argue that depreciation of environmental capital is equal to the costs associated with water, air, and noise pollution. Building on this, this paper calculates depreciation of environmental capital by using the growth rate of the cost of environmental pollution—waste water, exhaust gas, solid waste, and noise. The initial step involves calculating the total investment in water treatment, waste gas treatment, solid waste treatment, and noise control for each region and then uses the average annual growth rate of the total environment treatment in each region as the environmental depreciation.[10] Through this methodology, the depreciation rate of environmental capital is 0.127.

[10] A higher growth rate in environmental pollution control indicates increased pollution generation and greater environmental 'wear.' Maintaining the same environmental standard would necessitate proportionally higher investment in environmental capital, suggesting faster environmental capital depreciation.

Table 2
Capital investment resources.

Capital	Sector	Resource	Definition
Physical capital	Public sector Private sector	Fixed capital investment	The workload and related costs of constructing and purchasing fixed assets throughout society
Human capital	Public sector	Financial expenditure on education Financial medical and health expenditure	Government expenditure on education affairs Government expenditure on healthcare.
	Private sector	Education expenses Residents' health care expenditure	Income obtained from teaching and auxiliary activities carried out by higher education institutions The total cost of drugs, supplies, and services used by residents for medical and healthcare purposes
Environmental capital	Public sector	Financial expenditure on environmental protection Environmental infrastructure investment	Government environmental protection expenditure Expenses related to environmental infrastructure construction
Social capital	Public sector	Financial social security and employment expenditure Financial public security expenditure Financial culture and sports expenditure Financial transportation expenditure	Government expenditure on social security and employment Government expenditure on maintaining social and public safety Government expenditure on culture, cultural relics, sports, radio and television, news and publishing, etc. Government highway, waterway , railway , civil aviation transportation expenditure, etc
	Private sector	Residents' transportation and communication expenditure	Residents' expenses for transportation and communication tools and related expenses

Note: The data are from National Bureau of Statistics, China Education Statistics Yearbook, China Environment Statistics Yearbook and Statistical Yearbook by Province.

Social Capital: The role of social networks in economic growth depends on levels of trust. Only social networks capable of fostering widespread trust can effectively promote economic growth (Granovetter, 1985). Consequently, social capital can be measured investment in social networks, enhancement of social trust, and the establishment of social norms and laws that foster cooperation and transactions. This integrates three aspects: social networks, social trust, and social norms. Financial social security and employment expenditure, and financial public security expenditure strengthen social trust, creating a safer and more reliable social environment. Financial culture and sports expenditure, and financial transportation expenditure enhance the cohesion of social networks, fostering interpersonal connections among the population. Public security expenditure strengthens the adherence to laws and social conduct codes. For private sector investment, residents' transportation and communication are considered. The depreciation rate for social capital is set at 0.1, as suggested by Ishise and Sawada (2009).

The translog function is used to calculate the output elasticity of each capital in province i at time t:

$$\begin{aligned} \ln(y_{it}) = {} & \beta_0 + \beta_k \ln(k_{it}) + \beta_h \ln(h_{it}) + \beta_r \ln(r_{it}) + \beta_s \ln(s_{it}) + \beta_{kh} \ln(k_{it}) \times \ln(h_{it}) + \beta_{kr} \ln(k_{it}) \times \ln(r_{it}) \\ & + \beta_{ks} \ln(k_{it}) \times \ln(s_{it}) + \beta_{hr} \ln(h_{it}) \times \ln(r_{it}) + \beta_{hs} \ln(h_{it}) \times \ln(s_{it}) + \beta_{rs} \ln(r_{it}) \times \ln(s_{it}) + \beta_{tk} t \times \ln(k_{it}) \\ & + \beta_{th} t \times \ln(h_{it}) + \beta_{tr} t \times \ln(r_{it}) + \beta_{ts} t \times \ln(s_{it}) + \beta_T t + \beta_{TT} t^2 - u_{it} + v_{it} \end{aligned} \tag{23}$$

Output y is expressed in real GDP per capita. Let $z_{it} \in \{k_{it}, h_{it}, r_{it}, s_{it}\} = Z$ and $Z_- = Z \setminus z_{it}$. The output elasticity for each form of capital, z_{it}, is given by:

$$e_{z,it} = \frac{\partial ln(y_{it})}{\partial ln(z_{it})} = \beta_z + \sum_{v \in Z_-} \beta_{vz} \ln(v_{it}) + 2\beta_{zz} \ln(z_{it}) + \beta_{tz} t \tag{24}$$

The estimated average output elasticity from 2007 to 2020 is shown in Fig. 2.

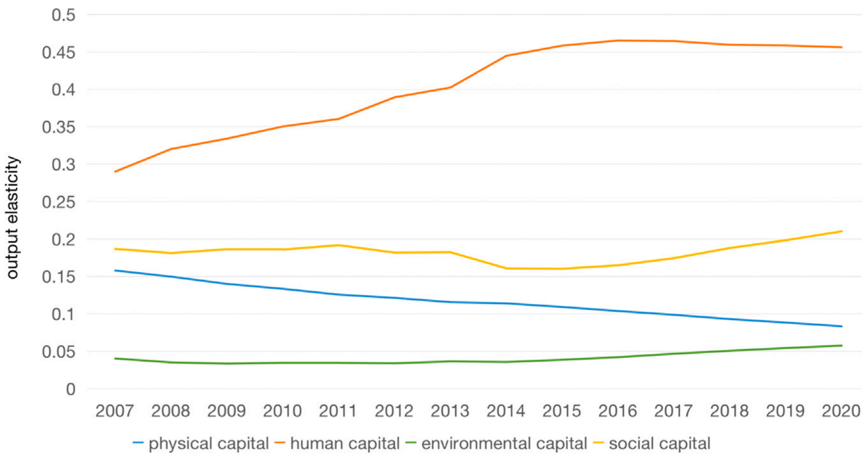

Fig. 2. Output elasticity of physical capital, human capital, environmental capital, social capital.

The data reveals that human capital has the highest output elasticity, followed by social capital, physical capital, and environmental capital. The average output elasticity for human capital, social capital, physical capital, and environmental capital are

0.408, 0.182, 0.116, and 0.041, respectively. The elevated output elasticity of human capital underscores its pivotal role in economic growth. Notably, this elasticity has risen from 0.29 in 2007 to 0.46 in 2020, emphasizing its growing importance over time. The elasticity of social capital remains relatively stable at around 0.18. Conversely, the elasticity of physical capital exhibits a declining trend, indicating diminishing returns from investments for economic growth. Although the output elasticity of environmental capital is the lowest, suggesting limited immediate returns on such investments, it is gradually increasing. This suggests that investment in environmental capital should not be readily overlooked, particularly for medium- to long-term planning.

4.2. Empirical analysis of capital structure on sustainable development (H1)

The dependent variable in this analysis is the sustainable development indicator, while the covariates of interest are the three degrees of deviation and a lag of the sustainable development indicator. The control variables used are the level of financial development (Hunjra et al., 2022), the level of foreign direct investment (FDI) (Aust et al., 2020), industrial structure (Zhu et al., 2019), urban-rural structure, and energy source structure (Vera and Langlois, 2007). The level of financial development is represented by the ratio of total deposits and loans to GDP. The level of FDI is quantified as the proportion of FDI to GDP. Industrial structure is defined as the ratio of the added value of the tertiary industry to GDP. Urban-rural structure is measured by the proportion of the urban population to the total population. Lastly, the energy structure is measured using ratio of coal consumption to total energy consumption.[11] Given that this paper employs a dynamic panel data model, estimation is conducted using the generalized method of moments (GMM).

$$\text{Sustainable\_Development}_{i,t} = \beta_0 + \beta_1 \text{Sustainable\_Development}_{i,t-1} + \beta_2 (D_{xy})_{i,t} + \sum_{j}^{n} \alpha_j \text{Controls}_{i,t} + u_i + v_t + \epsilon_{i,t} \tag{25}$$

for $x, y \in \{h, s, r\}$ and $x \neq y$, with i representing provinces and t representing time in years.

This paper undertakes three sets of empirical analyses, each focusing on the proximity of capital investment ratios to their respective elasticity ratios. The analyses employ system GMM, two-step GMM, and GMM on a sub-sample spanning 2007–2019 to mitigate the confounding impacts of the COVID-19 pandemic. Table 3 reveals that both AR2 and Hansen's test yield values greater than 0.1, validating the appropriateness of the GMM-based empirical analyses. For each degree of deviation – D_{hr}, D_{hs}, D_{sr} – a significant and negative correlation emerges, that is, lower degrees of deviation correspond to higher levels of sustainable development. This suggests that investment ratios closer to their respective elasticity ratios yield higher sustainable development levels, thereby empirically confirming the model's predictions i.e. H1.

The findings underscore the need for precise measurement of the output elasticity of various forms of capital. The gap between input ratios and output elasticity ratios serves as an indicator of the efficiency of resource allocation. Higher efficiency in allocating human, environmental, and social capital correlates with higher levels of sustainable development. Improved investment strategies are essential for achieving sustainable development, and the key to these strategies lies in maintaining balanced growth across various forms of capital.

Table 3
GMM estimations between deviation degree and sustainable development.

VARIABLES	(1) SYS-GMM	(2)	(3)	(4) DIF-GMM	(5)	(6)	(7) Year 2007–2019	(8)	(9)
Sustainable development (1 lag)	1.0710***	1.0688***	1.0680***	1.0034***	0.9598***	0.9950***	1.1170***	1.1163***	1.1107***
	(0.00)	(0.00)	(0.00)	(0.01)	(0.01)	(0.01)	(0.00)	(0.00)	(0.00)
D_{hr}	−0.0199***			−0.0081***			−0.0189***		
	(0.00)			(0.00)			(0.00)		
D_{hs}		−0.0143***			−0.0173***			−0.0143***	
		(0.00)			(0.00)			(0.00)	
D_{sr}			−0.0323***			−0.0059**			−0.0303***
			(0.01)			(0.00)			(0.00)
Financial development level	−0.0548	−0.0823**	−0.0462	−1.3582***	−1.6640***	−1.4531***	−0.1319***	−0.1470***	−0.1217***
	(0.04)	(0.03)	(0.04)	(0.06)	(0.11)	(0.14)	(0.03)	(0.03)	(0.04)
Foreign direct investment level	−16.0663***	−15.8327***	−15.4306***	−31.1072***	−30.5328***	−34.1924***	−12.7268***	−12.2264***	−12.0688***
	(2.20)	(1.96)	(2.50)	(2.22)	(3.20)	(4.00)	(1.24)	(0.95)	(1.21)
Industrial structure	0.5147***	0.5451***	0.5661***	3.1884***	4.0394***	3.8871***	0.5607***	0.5637***	0.6259***
	(0.06)	(0.06)	(0.06)	(0.20)	(0.15)	(0.24)	(0.06)	(0.05)	(0.05)
Urban-rural structure	1.4343***	1.9967***	1.3401***	1.3371	5.3788**	3.0067	0.1661	0.6594**	0.1233
	(0.36)	(0.32)	(0.37)	(1.60)	(2.16)	(2.21)	(0.28)	(0.26)	(0.30)
Energy resource structure	−0.1438	−0.1471	0.0108	−7.1205***	−5.9487***	−3.8517***	0.4423***	0.4739***	0.5535***
	(0.21)	(0.22)	(0.15)	(0.73)	(1.65)	(1.36)	(0.14)	(0.14)	(0.16)
Constant	−0.2377	−0.4953***	−0.3342***	4.7375***	2.5426	2.0647	−0.2359*	−0.5013***	−0.3295***
	(0.18)	(0.17)	(0.10)	(0.99)	(1.72)	(1.54)	(0.12)	(0.12)	(0.11)
Observations	390	390	390	390	390	390	360	360	360
Number of province	30	30	30	30	30	30	30	30	30
AR2	0.763	0.857	0.802	0.9803	0.8639	0.9509	0.221	0.198	0.209
Hansen/Sargan	0.194	0.191	0.190	1.0000	1.0000	1.0000	0.125	0.123	0.122

Note 1: Standard error in parentheses. ***,**,* denote statistical significance at the 1%, 5%, and 10% levels, respectively.
Note 2: In order to make the results more readable, the sustainable development indicator is multiplied by 100.

[11] Coal consumption is the sum of seven related energy terminal consumption, and total energy consumption is twenty related energy terminal consumption.

4.3. Empirical analysis of capital structure on consumption per capita (H2)

This paper uses the consumer price index (CPI) to calculate real consumption per capita. The control variables used are real GDP per capita, the retail price index (RPI), year-end deposits, the level of financial development, and industrial structure. Firstly, GDP per capita serves as an indicator of a province's overall economic development, which in turn influences per capita consumption. Secondly, fluctuations in the RPI either dampen or stimulate consumer demand, making it a relevant control variable for consumption. Thirdly, year-end deposits are used to gauge residents' propensity to save, which significantly impacts consumption. Fourthly, some studies suggest that financial development intensifies market competition, thereby reducing the cost of financial intermediation and promoting consumption (Jappell and Pagano, 1989). Lastly, shifts in industrial structure can influence consumption expenditure (Zhang et al., 2023).

This paper carries out three sets of empirical analyses based on the equation below, presenting results for (i) system GMM, (ii) GMM with autonomous regions removed, and (iii) GMM for a sub-sample of the dataset (2007–2019). The exclusion of autonomous regions is due to their local governments typically having greater autonomy in internal affairs compared to other provincial administrative regions, which could uniquely influence consumption patterns. The use of a sub-sample for the years 2007–2019 is for reasons previously outlined.

$$\text{Consumption}_{it} = \beta_0 + \beta_1 \text{Consumption}_{it(-1)} + \beta_2 \ln\left(\frac{k_{it}}{x_{it}}\right) + \sum_{j}^{n} \alpha_j \text{Controls}_{it} + u_t + v_t + \epsilon_{it} \tag{26}$$

for $x \in \{h, s, r\}$

The regression results are displayed in Table 4. The empirical findings reveal a significant negative correlation between k/h (k/r, k/s) and consumption, which is in alignment with the theoretical model. These results suggest that higher investment in human, environmental, and social capital, as opposed to physical capital, is associated with higher levels of consumption. Relatively speaking, the rate of return on investment for physical capital lags behind that of human and social capital. In summary, enhancing investment in human, environmental, and social capital – or reducing investment in physical capital – can effectively elevate consumption levels and thereby contribute to sustainable development.

Table 4
Non-linear GMM regression results of the ratio of k/h, k/r and k/s on consumption per capita.

VARIABLES	(1) SYS-GMM	(2)	(3)	(4) Exclude provinces	(5)	(6)	(7) Year 2007–2019	(8)	(9)
Consumption (1 lag)	1.2705***	1.2700***	1.2753***	1.2424***	1.2230***	1.2447***	1.1619***	1.1598***	1.1631***
	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)
Consumption (2 lag)	−0.2834***	−0.2880***	−0.2915***	−0.2591***	−0.2395***	−0.2593***	−0.2055***	−0.2087***	−0.2082***
	(0.02)	(0.01)	(0.01)	(0.02)	(0.03)	(0.02)	(0.02)	(0.02)	(0.02)
$ln(k/h)$	−0.0769**			−0.0940*			−0.1197***		
	(0.03)			(0.05)			(0.03)		
$ln(k/r)$		−0.0819***			−0.0943***			−0.0894***	
		(0.01)			(0.02)			(0.01)	
$ln(k/s)$			−0.0607***			−0.0820***			−0.0795***
			(0.02)			(0.03)			(0.02)
Real GDP per capita	0.2316***	0.2794***	0.2388***	0.2773***	0.3160***	0.2629***	0.4449***	0.4879***	0.4363***
	(0.04)	(0.04)	(0.05)	(0.05)	(0.05)	(0.05)	(0.06)	(0.06)	(0.06)
Year-end deposit	0.0011***	0.0018***	0.0011***	0.0008***	0.0017***	0.0010***	0.0015***	0.0024***	0.0016***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Retail price index	−0.0099***	−0.0109***	−0.0101***	−0.0131***	−0.0158***	−0.0133***	−0.0088***	−0.0097***	−0.0087***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Financial development level	0.0176	0.0172	0.0183	0.0241	0.0081	0.0074	0.0641***	0.0679***	0.0640***
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Industrial structure	0.0368***	0.0366**	0.0499***	0.0426***	0.0392**	0.0526***	0.0388**	0.0390	0.0533***
	(0.01)	(0.02)	(0.01)	(0.01)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Constant	−0.7251*	−0.9758**	−0.8159	−0.7989	−0.7515*	−0.6456	−2.7356***	−3.0229***	−2.7596***
	(0.39)	(0.38)	(0.49)	(0.48)	(0.43)	(0.54)	(0.55)	(0.55)	(0.57)
Observations	360	360	360	312	312	312	330	330	330
Number of province	30	30	30	26	26	26	30	30	30
AR2	0.780	0.795	0.819	0.943	0.950	0.935	0.300	0.285	0.298
Hansen	0.228	0.210	0.220	0.417	0.392	0.408	0.165	0.122	0.164

Note: Standard error in parentheses. ***,**,* denote statistical significance at the 1%, 5%, and 10% levels, respectively.

5. Conclusion

This study explores the impact of targeted investments in various forms of capital, beyond physical capital, on sustainable development. A new theoretical model on sustainable development is established which extends the standard welfare function approach. Based on the predictions of the model, the empirical findings yield several key insights:

Firstly, when the investment ratios in human, environmental, and social capital align closely with their respective output elasticity ratios, sustainable development levels are higher. This means that when the ratio between any two forms of capital more closely matches the ratio of their respective output elasticities, the level of sustainable development approaches its maximal level. This insight is pivotal for understanding how China's capital structure can evolve to better support sustainable development.

Secondly, a negative relationship exists between the ratio of physical capital to human/ environmental/social capital and per capita consumption. During times of elevated per capita consumption, achieving sustainable development necessitates a greater focus on investments in human, environmental, and social capital, as opposed to physical capital.

Thirdly, human capital consistently exhibits higher output elasticity compared to physical, social, and environmental capital. This suggests that as China's economic structure and development evolve, human capital is assuming an increasingly vital role in sustainable economic development. Moreover, the output elasticity of environmental capital is on the rise, promising further positive externalities in the medium to long term and thereby reinforcing the cycle of sustainable development. In this context, this paper argues that environmental investment warrants even greater attention than it has historically received.

The findings carry substantial policy implications. Accurate estimation of the output elasticities of various forms of capital is crucial. Government investment strategies should be guided by these (relative) output elasticities to ensure effective allocation of resources. To pursue sustainable development, given the diminishing returns from physical capital accumulation, the focus should shift toward human, environmental, and social capital

For subsequent studies, the current model focuses on four types of capital but could be expanded to include financial capital, creating a five-capital framework. Another avenue to explore is the structural relationship between public and private capital investment to better inform government policy. Lastly, while the model's conclusions are currently applicable only to China, extending the research to include other countries remains a valuable goal.

CRediT authorship contribution statement

Yiyang Cheng: Conceptualization, Methodology, Software, Data curation, Writing – original draft. **Shaofeng Ru:** Funding acquisition, Writing – original draft, Supervision. **Gavin Kader:** Visualization, Investigation, Supervision, Reviewing & editing.

Declaration of competing interest

There is no conflict of interest in the submission of this manuscript.

Data availability

Data will be made available on request.

Acknowledgements

The authors acknowledge the editor and anonymous reviewers for their valuable comments and suggestions. This research is funded by Chinese National Funding of Social Sciences (No. 19AJL010) and National Natural Science Foundation of China (No. 72033006).

Appendix

A.1. Optimal path derivation process

$$\frac{\partial H}{\partial \overline{c}} = \overline{c}^{-\theta} - \lambda + z = 0 \tag{27}$$

$$\frac{\partial H}{\partial I_H^g} = \frac{\partial H}{\partial I_H^p} = \lambda - m = 0 \tag{28}$$

$$\frac{\partial H}{\partial I_R^g} = \lambda - n = 0 \tag{29}$$

$$\frac{\partial H}{\partial I_S^g} = \frac{\partial H}{\partial I_S^p} = \lambda - l = 0 \tag{30}$$

$$\frac{\partial H}{\partial \overline{k}} = \lambda A \alpha k^{\alpha-1} h^{\beta} r^{\gamma} s^{1-\alpha-\beta-\gamma} = \rho\lambda - \dot{\lambda} \tag{31}$$

$$\frac{\partial H}{\partial \overline{h}} = \lambda A \beta k^{\alpha} h^{\beta-1} r^{\gamma} s^{1-\alpha-\beta-\gamma} + \omega = \rho m - \dot{m} \tag{32}$$

$$\frac{\partial H}{\partial \overline{r}} = \lambda A \gamma k^{\alpha} h^{\beta} r^{\gamma-1} s^{1-\alpha-\beta-\gamma} + \omega = \rho n - \dot{n} \tag{33}$$

$$\frac{\partial H}{\partial \overline{s}} = \lambda A (1-\alpha-\beta-\gamma) k^{\alpha} h^{\beta} r^{\gamma} s^{1-\alpha-\beta-\gamma} + \omega = \rho l - \dot{l} \tag{34}$$

$$z(\overline{c} - \overline{c}_{min}) = 0 \tag{35}$$

To make Eq. (35) hold, z has to be 0. From Eqs. (27)–(30), $\overline{c}^{-\theta} = \lambda = m = n = l$. Substituting into Eq. (31)–(34) yields the two necessary conditions in 3.3.

Table 5
Descriptive statistical results.

Variable	Obs	Mean	Std. dev.	Min	Max	Variable	Obs	Mean	Std. dev.	Min	Max
Sustainable development	420	13.05939	9.714349	3.65558	79.95184	Consumption	420	10.32453	5.244982	3.1839	31.6052
Dhr	420	4.279924	8.032372	0.0019028	82.5131	ln(k/h)	420	1.977236	0.3035746	0.8227	2.806
Dhs	420	4.662733	9.033388	0.001082	67.59782	ln(k/r)	420	3.291255	0.757914	0.2689	4.3882
Dsr	420	2.019829	4.197932	0.0100034	40.59915	ln(k/s)	420	1.905291	0.3428665	0.9008	2.6987
Financial development level	420	3.051188	1.163951	1.392	8.131	Year-end deposit	420	38.21174	39.46918	1.09	267.64
Foreign direct investment level	420	0.0211193	0.0196373	0.0001074	0.1209925	Real GDP per capita	420	10.19929	0.5595772	8.694659	11.60155
Industrial Structure	420	1.120965	0.6527163	0.499599	5.29682	Retail price index	420	102.1035	2.029283	96.82	110.56
Urban-rural structure	420	0.5585048	0.1337947	0.283	0.946						
Energy resource structure	420	0.4121039	0.1536816	0.0071068	0.724146						

Table 6
Correlation matrix for 4.3.

	Sustainable development	Dhr	Dhs	Dsr	Financial development level	Foreign direct investment level	Industrial Structure	Urban rural structure	Energy resource structure
Sustainable development	1								
Dhr	−0.0293	1							
Dhs	0.1594	0.2378	1						
Dsr	−0.159	0.6163	−0.0679	1					
Financial development level	0.7662	−0.0348	0.0993	−0.1043	1				
Foreign direct investment level	0.0505	0.0342	0.2262	0.0559	−0.0377	1			
Industrial structure	0.8128	−0.0771	0.04	−0.1119	0.8164	0.016	1		
Urban-rural structure	0.7218	−0.0453	0.3422	−0.2552	0.6212	0.3174	0.5557	1	
Energy resource structure	−0.6493	0.0082	−0.2118	0.1925	−0.6277	−0.1124	−0.6129	−0.6192	1

Table 7
VIF for 4.3.

Variable	VIF	1/VIF	Variable	VIF	1/VIF	Variable	VIF	1/VIF
Financial development level	3.74	0.267502	Financial development level	3.73	0.268347	Financial development level	3.78	0.264475
Industrial Structure	3.19	0.313547	Industrial Structure	3.22	0.310116	Industrial Structure	3.16	0.316161
Urban-rural structure	2.28	0.439495	Urban-rural structure	2.44	0.410194	Urban-rural structure	2.43	0.411928
Energy resource structure	2.01	0.498614	Energy resource structure	2.02	0.49548	Energy resource structure	2.01	0.49657
Foreign direct investment level	1.24	0.805877	Foreign direct investment level	1.25	0.801112	Foreign direct investment level	1.28	0.780996
Dhr	1.01	0.985955	Dhs	1.2	0.835063	Dsr	1.12	0.892848
Mean VIF	2.24		Mean VIF	2.31		Mean VIF	2.3	

Table 8
Correlation matrix for 4.4.

	Consumption	ln(k/h)	ln(k/r)	ln(k/s)	Real GDP per capita	Year-end deposit	Retail price index	Financial development level	Industrial structure
Consumption	1								
ln(k/h)	−0.1071	1							
ln(k/r)	0.4037	0.4261	1						
ln(k/s)	−0.1459	0.8917	0.4808	1					
Real GDP per capita	0.9014	0.1849	0.5613	0.1161	1				
Year-end deposit	0.745	−0.1191	0.4476	−0.1314	0.6519	1			
Retail price index	−0.3596	−0.0973	−0.3972	−0.1141	−0.3845	−0.2627	1		
Financial development level	0.7291	−0.4353	0.1041	−0.4651	0.5244	0.48	−0.3283	1	
Industrial structure	0.6881	−0.3972	0.0647	−0.3642	0.4862	0.4486	−0.2957	0.8164	1

A.2. Descriptive statistical results

See Table 5.

A.3. Correlation matrix and VIF results

See Tables 6–9.

Table 9
VIF for 4.4.

Variable VIF	VIF	1/VIF	Variable	VIF	1/VIF	Variable	VIF	1/VIF
Financial development level	3.85	0.259609	Financial development level	3.33	0.300337	Financial development level	4.14	0.241391
Industrial structure	3.14	0.318157	Industrial structure	3.14	0.31893	Industrial structure	3.05	0.327372
Real GDP per capita	2.96	0.3383	Real GDP per capita	2.54	0.393624	Real GDP per capita	2.6	0.385216
ln(k/h)	1.96	0.50892	Year-end deposit	1.93	0.518733	Year-end deposit	1.89	0.528013
Year-end deposit	1.94	0.515025	ln(k/r)	1.86	0.538151	ln(k/s)	1.81	0.553355
Retail price index	1.25	0.802664	Retail price index	1.34	0.748544	Retail price index	1.27	0.785908
Mean VIF	2.52		Mean VIF	2.35		Mean VIF	2.46	

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